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Anaerobic digestion of corn straw pretreated by ultrasonic combined with aerobic hydrolysis

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HIGHLIGHTS

- An effective way was studied for corn straw fermentation experiment.
- The lag phase of fermentation can accelerate after ultrasonic pretreatment.
- The performance of AD and the yield of methane can improve by ultrasonic pretreatment.
- The highest cumulative methane production was 198.56 mL/g VS, 46.73% higher than untreated group.

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ABSTRACT

Corn straw (CS) was pretreated by ultrasonic combined aerobic with biogas slurry as medium for anaerobic digestion (AD), that strengthened the degradation efficiency CS, varied in the composition of digestion slurry, thereby the methane production was increased. Central combinatorial design (CCD) test was used to treat CS at ultrasonic power (200, 400, and 600 W), time (10, 20, and 30 min) and AD for 25 days, at 37 ± 1℃. According to data showed that the pH and volatile fatty acids (VFAs) affected methane production directly. With an ultrasonic power 309 W, time 26 min, it reached the maximum content of VFAs with 16.24 g/L, the cumulative methane production achieved the highest with 198.56 mL/g VS, which was 46.73% higher than unpretreated raw material as CK. Ultrasonic-aerobic hydrolysis pretreatment can obtain higher VFAs and methane production content in a short period of time that is great significance to biogas engineering.

1. Introduction

Since 2014, China has made great efforts to solve the adverse impact of corn straw (CS), and proposed that taking advantage of CS for AD to produce methane can not only solve the problem of CS, but also provide energy (Chen et al., 2019; Zhang et al., 2019). The use of AD to transform biomass raw materials into biomass energy has been the most widely resource utilization way in China, meanwhile, the products of reaction are also widely used to achieve zero pollution (Lorena, et al., 2018; Sun et al., 2020). However, due to the complex structure of cell walls of CS, anaerobic microorganisms cannot directly break through the lignocellulosic construction to utilize, so it is vital to apply pretreatment ways to improve the efficiency of AD (Liu et al., 2017). Up to now, the common pretreatment methods mainly include physical, chemical, biological, ultrasonic pretreatment, steam explosion and liquid hot water pretreatment (Zhong et al., 2018; Syaftika and Matsumura, 2018; Zou et al., 2016; Li et al., 2017).

For instance, Wang et al. (2019) found that the pretreatment of CS with production wastewater from furfural plant could increased the content of volatile fatty acids (VFAs) in the substrate and effectively

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Table 1
Characteristic of the digestion substrate.

Index	CS	Inoculum
Total Solids (%)	92.52 ± 0.02	3.58 ± 0.01
Volatile Solids (%)	89.54 ± 0.04	1.21 ± 0.02
pH	–	7.04 ± 0.03
Total Carbon (%)	40.95 ± 0.03	32.01 ± 0.01
Total Nitrogen (%)	0.49 ± 0.06	3.56 ± 0.03
Cellulose (%)	35.59 ± 0.52	–
Hemicellulose (%)	28.06 ± 0.34	–
Lignin (%)	8.65 ± 0.32	–

Notes: "%"– dry matter mass, "-"–not determined.

increased cumulative methane production. Kang et al. (2018) reported that hemicellulose and lignin were removed by alkali pretreated and the yield of methane production increased by 21%. Some researcher reported that used efficient white rot fungus strains to treat CS to improve the porosity and crystallinity of lignocellulose in AD process (Rouches et al. 2017; Zhao et al., 2020; Chu et al. 2020). Above mentioned acid, alkali and biological pretreated methods, usually purported expensive cost prior to AD. Therefore, it is urgent to find an efficient and pollution-free method to preprocess CS before AD. Ultrasonic pretreatment was proposed the most popular technologies to accelerate the biochemical process of microorganisms, recently (Yang and Fang, 2014; Zheng et al., 2014), with advantages of easy operation and good effect in producing gas. The ultrasonic pretreated method has been mature and applied in many large and medium-sized methane projects gradually (Batghare et al., 2018).

A large number of studies showed that ultrasonic technology could rupture and deform fiber cells, or break the hydrogen bonds between fiber molecules, and increase the inner surface area. (Hassan, et al., 2018; Subhedar and Gogate, 2014). Fakayode et al. (2020) used ultrasonic and deep eutectic solvent pretreatment to continuously removed 43.56% of lignin from watermelon, at ultrasonic power of 180 W. Bussemaker et al. (2013) found that the low-frequency ultrasound could result in the deploy-merization of lignin matrix through the cleavage of α-o-4 and β-o-4 bonds, and free radicals also caused the rupture of lignocarbohydrate bond, and releasing hemicellulose. The United States, United Kingdom, Australia and other countries have begun to use ultrasonic technology to treat waste activated sludge. Chen et al. (2014) investigated that it could effectively reduce the volatile solid (VS) organics and increase biogas production by more than 40% on average. Therefore, it is very desirable that use the ultrasonic method to promote the disintegration of CS (Srisowmeya et al., 2020). The introduction of aerobic hydrolysis in the process of AD and the establishment of a unique micro-oxygen environment has a positive impact on microbial metabolism (Nguyen and Khanal, 2018; Nguyen et al., 2019). In addition, the addition of low-oxygen micro-aerobic conditions in the anaerobic digester can significantly increase the degradation rate of lignocellulose and the methane production in the anaerobic process (Fu et al., 2015; Xu et al., 2018).

The main purposes of this research are: (1) to inquire into the changes of pretreatment slurry at synergistic effect of ultrasonic and oxygen hydrolysis; (2) to analyze lignocellulosic degradation rate, reducing sugar content, VFAs, pH and cumulative methane production of CS in AD; and (3) to use the response surface optimization analysis to find the optimum proposal for improving VFAs. This research provides an economical and practical way for the pretreatment of complex raw materials during the operation of biogas projects.

2. Materials and methods

2.1. Digestion substrate

After natural air drying, the CS was crushed into 0.5–1 cm pieces and stored for use later. Inoculum and CS were cultivated in the laboratory of Northeast Agricultural University in Harbin, China. The characteristics of the inoculum and CS are presented in Table 1. All other chemicals and solvents are analytical grade.

Table 2
Code and levels of experimental variables.

Code value	Factors	units	Level of coding −1.414	−1	0	+1	+1.414
A	Power	W	200	258.57 (259)	400	541.42 (541)	600
B	Time	min	10	12.93 (13)	20	27.07 (27)	30

2.2. Experiment design

Accurately weigh 10 g of raw material and CS by ultrasonic treated and put them in 100 mesh cotton filter bags, seal and press into the biogas slurry, keep the water bath at 37 °C for 0.5 h, 1.0 h, 2 h, 4 h, 6 h, 8 h, 12 h and 16 h, thereby take it out, rinse off the impurities with water, drain for 5 min until no more dripping, measure moisture content.

The test is divided into two parts: ultrasonic combined aerobic hydrolysis pretreatment and AD. The ultrasonic pretreatment device used in this experiment was designed and developed by the Chemistry Laboratory of Northeast Agricultural University. The ultrasonic control device, ultrasonic generator and acoustic chamber together form the entire ultrasonic pretreatment device. The ultrasonic control apparatus including the power, duration time and temperature. The ultrasonic generator consists of an immersion probe with a diameter of 1.5 cm and a length of 19 cm. They can emit ultrasonic waves in both horizontal (X-axis) and vertical (Z-axis) directions. The ultrasonic frequency of the device is 20 kHz, and the maximum power of the device is 900 W.

Adjust the lug of the lifting platform so that the transmitting end of the ultrasonic probe is immersed in the solution of 10–30 mm of the liquid level (working volume: 200 mL). 40 g CS (moisture content 11.58%) and 400 mL biogas slurry with TS of 5% were mixed in 1000 mL beaker, and placed them under the generator for ultrasonic combined aerobic hydrolysis pretreatment, for the different ultrasonic power and time to ger pretreatment slurry, at 25℃. The experiment was divided into two groups with 3 parallels, one was used for the detection of the slurry composition after ultrasonic pretreatment, and the other was all transferred to the subsequent AD. The experimental conditions of the two groups are exactly the same. They were carried out in a constant temperature water bath at 37 °C for 25 days, with regular shaking once a day to keep the materials uniform. 40 g unpretreated CS and 400 mL biogas slurry were used as substrate of CK, and adjust the substrate TS to 5% for AD immediately.

2.3. Central composite design

CCD test was used to explore the response of power and time of the ultrasonic-aerobic hydrolysis pretreatment to the VFAs index, and the second-order RSM was usually applied to fitting to optimize the arguments of ultrasonic-aerobic hydrolysis pretreatment conditions. Among them, power, time of two independent variables are respectively turned into coded A, B and divided into 5 levels: -λ, −1, 0, 1, λ. On the basis of previous study, in this study determined the range of independent variables in the CCD test. The encoding values of the independent variables and the actual values of the preprocessing conditions are shown in Table 2. At the same time, a multiple linear regression model was selected to analyze the response of ultrasonic-aerobic hydrolysis pretreatment to the content of VFAs as shown in Eq. (1):

$$Y = \beta_0 + \beta_1 A + \beta_2 B + \beta_{12} AB + \beta_{11} A^2 + \beta_{22} B^2 \quad (1)$$

Among them, Y is variable, representing VFAs; A is code, representing power; B is code, representing Time;β_0,β_1,β_2 are the model

Table 3
Factor level table.

Test groups	Experimental factor A(W)	B(min)
CK: Raw material	0	0
C1:200 W-20 min	200	20
C2:541 W-13 min	541	13
C3:259 W-27 min	259	27
C4:541 W-27 min	541	27
C5:259 W-13 min	259	13
C6:600 W-20 min	600	20
C7:400 W-10 min	400	10
C8:400 W-30 min	400	30
C9:400 W-20 min	400	20

coefficients of linear expressions; β_{11},β_{22} are coefficients, usually represent interaction between factors. According to the design expert software design experiment group, different CS processing conditions are shown in Table 3, and each experiment is set to 3 parallels.

2.4. Index measurement method

The samples are measured with differential gravimetric method for TS, VS and pH (pHS-25). Under standard atmospheric pressure, determine the water absorption rate of CS. Elemental analyzer is used for the determination of carbon and nitrogen (EA 3000, LEEMAN). The lignocellulose is measured using a Van Soest fiber analyzer (Ankom200i, Anko Borui Technology Beijing, China). The composition of biogas and VFAs are analyzed using different modules of gas chromatograph (GC-6890 N, Agilent Inc.).

2.5. Data statistical and analysis

The degradability of Lignocellulose of CS is determined by Van method (Van et al., 1991), which was described as Eq. (2):

$$Degradability\% = [1 - \frac{(P_j)_r M_r}{(P_j)_i M_i}] \times 100\% \quad (2)$$

Among them, M stands for total dry mass substance; i and r respectively represents the initial dry matter mass and remaining dry mass after treatment; (Pj) is percentage of various ingredients for M; j indicates different components such as lignin, cellulose and hemicellulose of CS.

The water absorption rate of CS is measured by the ratio of dry CS absorbing water solution, and eq is shown in (3):

$$w = \frac{(m - m_0)}{m_0} \times 100\% \quad (3)$$

Among them, w is water absorption rate of CS (%); m_0 is dry matter of CS (g); m is quality of CS saturate with water (g).

By fitting experimental data, the modified Gompertz equation can accurately predict methane production immediately (Wang et al., 2019), was described as Eq. (4):

$$Y(t) = Y_m exp\left\{ - exp\left[\frac{R_m e}{Y_m}(\lambda - t) + 1\right]\right\} \quad (4)$$

Among them, $Y(t)$ is cumulative methane production ($mL/g \quad VS$) at t

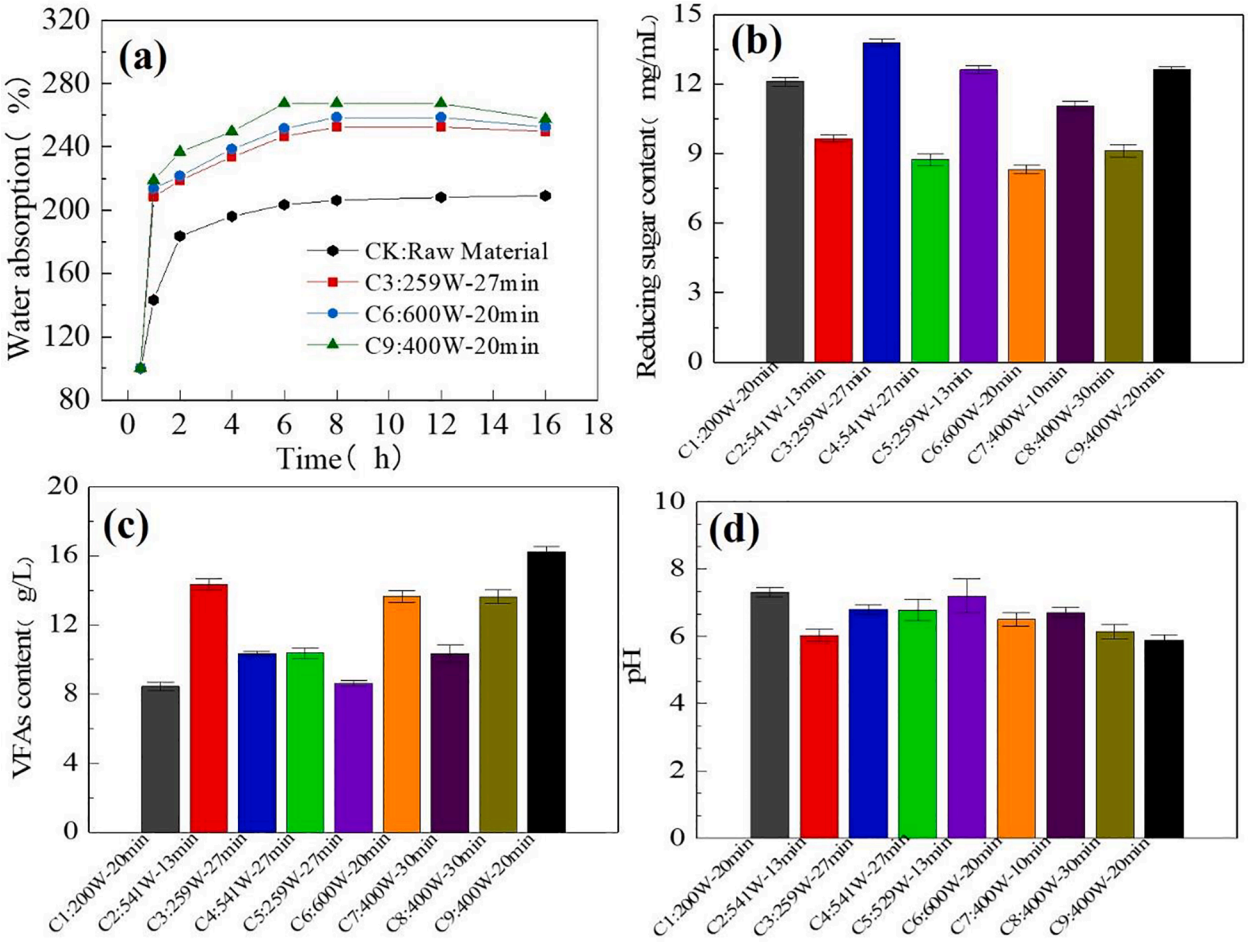


Fig. 1. Variation of parameters with ultrasound - aerobic hydrolysis; (a) water absorption, (b) Reducing sugar, (c) Volatile fatty acids content and (d) pH.

Table 4
Changes of CS composition by combine pretreatment.

Treatment	InitialQuality (g)	Quality after treatment(g)	Degradation rate of hemicellulose (%)	Degradation rate of cellulose(%)	Degradation rate of lignin (%)
C1:200 W-20 min	22.92	21.17 ± 0.15	28.29	15.58	3.52
C2:541 W-13 min	22.55	20.89 ± 0.08	36.00	18.91	3.98
C3:259 W-27 min	22.66	20.34 ± 0.72	31.35	16.89	4.64
C4:541 W-27 min	22.58	21.12 ± 0.79	32.76	19.30	6.35
C5:259 W-13 min	22.01	19.49 ± 0.70	28.98	24.69	6.94
C6:600 W-20 min	22.70	20.18 ± 0.31	36.71	28.82	7.83
C7:400 W-10 min	22.86	18.17 ± 0.26	29.60	19.65	6.54
C8:400 W-30 min	23.02	22.42 ± 1.01	36.56	26.33	5.60
C9:400 W-20 min	22.72	18.14 ± 0.53	38.51	30.17	8.69

(d); R_m is on behalf of maximal rate of methane production; λ proves lag phase time (d); Y_m explains usually potential maximal yield of methane production ($mL/g\ \ VS$); e≈2.718.

3. Results and discussion

3.1. Effect of pretreatment on the physical and chemical properties of CS

3.1.1. Effect of pretreatment on water absorption rate of CS

In this experiment, the C3:259 W-27 min, C6:600 W-20 min and C9:400 W-20 min groups were selected to determine the water absorption of CS and compared with CK. Fig. 1 (a) showed that the water absorption rate of CS increased with time before 6 h, and then remained unchanged after increasing to a certain value. The higher water absorption rate of CK, C3:259 W-27 min, C6:600 W-20 min and C9:400 W-20 min were 209.15%, 252.62%, 258.68% and 267.44%, respectively. The treatment groups showed better water absorption capacity, due to the increase in the pore density of CS of the cellulose skeleton at the conditions of ultrasonic-aerobic hydrolysis treatment, and the water molecules can get into the CS quickly (Dong and Chen, 2019; Velmurugan and Muthukumar, 2012). In the mid-term, the water absorption strength of CS decreased, because of the water in the CS was saturated and the diffusion rate decreased. The water absorption rate of CS decreased after 12 h, because the increase of cell wall rupture led to the decrease of water retention performance of CS. In summary, a certain degree of ultrasonic-aerobic hydrolysis pretreatment could expose more cellulose, expand the pore density of CS, and showed higher water absorption.

3.1.2. Effect of pretreatment on lignocellulosic of CS

The physical structure of CS treated by ultrasonic directly reflects the availability of this pretreatment method. Hence, it is very vital to quantitatively analyze and evaluate the degradation of lignocellulosic substances in CS (Guan et al., 2018). It can be seen from Table 4 that hemicellulose exhibited stronger degradability, with a degradation rate ranging from 28.29% to 38.51%. Secondly, the degradability of cellulose was 15.58%-30.17%, and the degradability of lignin was the weakness, which was 3.52%–8.69%. It was possible that ultrasonic cavitation broken hydrogen bonds, glycoside bonds in hemicellulose and showed higher degradability, enhanced CS solubilization (Kim et al., 2018; Qi et al., 2021). Due to the intensity of cavitation bubble effect loosened lignocellulose structure, exposed a larger number of enzyme action sites, which was more conducive to the degradation of lignocellulose (Rehman et al., 2014).

3.1.3. Scanning electron microscope analysis

Observed the CS of C9:400 W-20 min and CK through a scanning electron microscope 2000 times, we found that the surface of the wood fiber of CK was relatively smooth. Because there was no obvious holes in the surface of the vertical wood fiber, it is difficult for the hydrolase to enter the CS. The results of C9:400 W-20 min showed obvious holes and many “fragments”, as same time, the cellulose was stripped, and a sense of layering appeared. Ultrasonic-aerobic hydrolysis treatment significantly changed the structure of CS, and the surface of CS collapsed, indicating that the lignin removal effect was obvious. Compared with the ultrasonic-aerobic hydrolysis pretreatment, the CS has no obvious changes in the macroscopic view. However, under the electron microscope lens, the structure of the CS has changed significantly. Due to the effect of hydrolytic enzymes on the lignocellulose, and part of the hemicellulose and cellulose were degraded (Guan et al., 2018).

3.2. Effect of ultrasonic - aerobic hydrolysis on digestion slurry

3.2.1. Effect of ultrasonic - aerobic hydrolysis on reducing sugars

The degree of ultrasonic-aerobic hydrolysis on the hydrolysis and utilization of lignocellulose in CS, ultimately affect the performance of subsequent AD to produce methane (Hu et al., 2017). Variations of reducing sugar yields in pretreatment slurry directly reflected the performance of the hydrolysis stage under different ultrasonic-aerobic hydrolysis, was showed in Fig. 1(b). In all pretreated samples, the reducing sugar yields of pretreated C1:200 W-20 min, C2:541 W-13 min…C9:400 W-20 min varied among 8.31–13.79 mg/mL, which the highest one was C3 at ultrasonic power of 259 W and 27 min; the lowest one was C6 at ultrasonic power of 600 W, time in 20 min. Under the action of ultrasonic-aerobic hydrolysis, polar molecules contained in the CS could rub against each other violently, made the biomass surface with increased affinity for enzyme attack (Hu et al., 2017). Hence, the hydrolase was easily to enter the binding site of hydrolase to catalyze, accelerated hydrolyzed cellulose and hemicellulose into educing sugars such as glucose and xylose, provided abundant digestive substrate and carbon source for AD (Zou et al., 2016; Qi et al., 2020). However, excessive ultrasonic power would accelerate the degradation of reducing sugars, and decrease of pH in the digestion environment, which was not conducive to the survival of microorganisms and hydrolysis activities, and impact to the stability of hydrolysis system.

3.2.2. Effect of ultrasonic aerobic hydrolysis on VFAs and pH

Since methanogens cannot directly decompose and utilize macromolecular organic matter, but use VFAs such as acetic acid to convert into methane gas. Consequently, VFAs has become a key intermediate in the process of converting organic matter to methane (Liu et al., 2019). Variations of VFAs during the pretreated groups were shown in Fig. 1(c). The results displayed that the highest VFAs reached 16.24 g/L in C9:400 W-20 min; the lowest one was 8.45 g/L in C5:259 W-13 min. Ultrasonic pretreatment can make the carbohydrates are consumed and utilized by acid-producing bacteria into organic acids faster, thereby the VFAs increased in pretreated slurry (Lizama et al., 2017). As showed in Fig. 1 (d), the pH value all maintain generally between- 5.89–7.32. Due to the accumulation of VFAs lead to the pH drop drastically, the higher VFAs is contrary the lower pH value. Among the all pretreatments, the components of VFAs all showed the main VFAs in each treated groups were acetic acid, followed by butyric acid, propionic acid, and a very small amount of valeric acid. The VFAs can be used by acidogenic and acetogenic bacteria convert VFAs into acetate, H_2, and CO_2, which are used

Table 5
Study on the kinetics and actual parameters of AD cumulative methane production under different pretreatment conditions.

Test groups	Cumulative methane production kinetic parameters				Y_a(mL g^{-1} VS)
	Y_m(mL g^{-1} VS)	R_m(mL d^{-1} g^{-1} VS)	λ(d)	R^2	
CK: Raw material	134.26 ± 0.91	17.66 ± 0.59	1.37 ± 0.22	0.9885	135.32 ± 3.61
C1:200 W-20 min	152.68 ± 1.76	16.93 ± 1.10	0.24 ± 0.20	0.9855	154.32 ± 1.89
C2:541 W-13 min	182.61 ± 1.13	21.60 ± 0.77	0.39 ± 0.18	0.9967	184.03 ± 4.63
C3:259 W-27 min	174.52 ± 0.55	17.48 ± 0.63	0.36 ± 0.27	0.9965	174.32 ± 3.72
C4:541 W-27 min	175.36 ± 0.77	21.17 ± 1.00	0.18 ± 0.19	0.9985	176.78 ± 2.15
C5:259 W-13 min	167.93 ± 0.74	26.62 ± 1.50	0.46 ± 0.05	0.9993	168.20 ± 2.65
C6:600 W-20 min	193.18 ± 1.08	26.50 ± 0.94	0.66 ± 0.11	0.9879	194.36 ± 3.72
C7:400 W-10 min	179.64 ± 1.79	20.94 ± 1.16	0.23 ± 0.30	0.9907	180.71 ± 4.83
C8:400 W-30 min	187.71 ± 0.88	20.34 ± 1.15	0.58 ± 0.39	0.9933	191.34 ± 2.15
C9:400 W-20 min	196.07 ± 1.04	20.02 ± 0.91	0.28 ± 0.25	0.9945	196.13 ± 3.36

by the methanogens, make the reaction proceed toward the direction of methane production (Rashid and Li, 2020). Different VFAs components are utilized by one or more different bacteria, which are correspond to decomposition product, and affect the entire AD system together (Liu et al., 2018).

3.3. Effect of ultrasonic-aerobic hydrolysis on methane production in AD

The methane production of CS treated by ultrasonic-aerobic hydrolysis are shown in Table 5. The lowest and the highest treated groups are C1:200 W-20 min and C9:400 W-20 min, which are 154.32 mL/g VS and 196.13 mL/g VS, respectively. Compared with CK group, it has increased by 14.04%-52.33%. The ultrasonic-aerobic hydrolysis make the monosaccharides, and oligosaccharides are more readily water-soluble, and decompose by microorganisms and fungi quickly, and increase the initial VFAs content in the digestion slurry, thereby improving the methanogenesis characteristics of CS in AD. Among them, the fitting coefficient R^2 is greater than 0.98, indicating that the fitting effect is better (Yang et al., 2019). It can be seen that as the strength of aerobic hydrolysis increases, the lag time becomes shorter. The λ values of CK, C1:200 W-20 min, C2:541 W-13 min,…C9:400 W-20 min are: 1.37, 0.24, 0.39, 0.36, 0.18, 0.46, 0.66, 0.23, 0.58 and 0.28. The λ values of treatment groups were shorter than CK, significantly.

This relatively shorter lag phase indicated that the activity of methanogens is high at moderate pH, adapt the substrate faster, shortens the lag time of AD, resulting in a higher methane yield (Mainardis et al., 2019; Pellera and Gidarakos, 2016). Fig. 3 is a graph of cumulative methane production content, and Gompertz fitting was performed by origin 2017, and the fitting results are shown in the figure. The cumulative methane content of each experimental group is different, the cumulative methane production of the treated groups are higher than CK significant, and the difference between the groups was not significant. The above results showed that the pretreatment could effectively increase the cumulative methane production content (Y_a) of AD, which is similar to the fitting data conclusion (Y_m), indicating the fitting effect better. The main reason is that the hydrolase has improved cellulose degradation ability in the digestion system, and increased the VFAs in the initial stage of digestion. The ability of hydrogen and methanogens to VFAs has also been improved, and better treatment effect.

Table 6
Variance analysis of volatile fatty acids.

Source	Sum ofSquares	df	MeanSquare	F	PProb > F	
Mode	110.08	5	22.02	21.93	0.0004	Significant
A-Power	22.18	1	22.18	22.10	0.0022	
B-Time	0.71	1	0.71	0.71	0.4282	
AB	8.51	1	8.15	8.12	0.0247	
A^2	53.42	1	53.42	53.23	0.0002	
B^2	35.56	1	35.56	35.52	0.0006	
Residual	7.03	7	1			
Lack of fit	7.03	3	2.34			
Pure error	0.00	4	0.00			
Total	117.10	12				

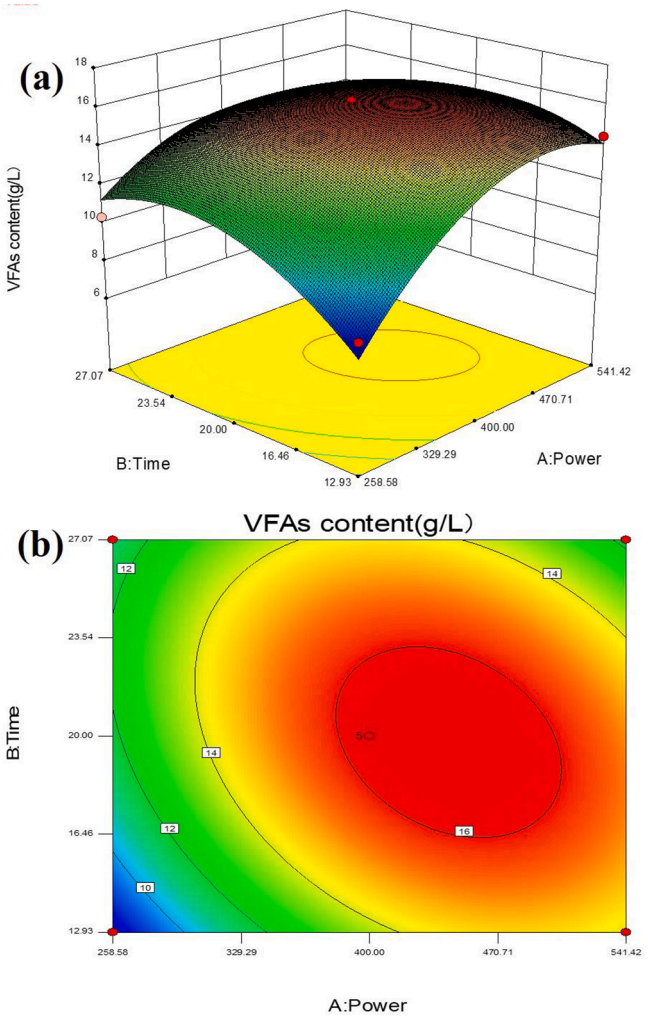


Fig. 2. (a) 3D surface and (b) Contour.

3.4. Response surface optimization parameters

In order to obtain the highest volatile fatty acid content, the conditions of ultrasonic-aerobic hydrolysis pretreatment were optimized by RSM method. Table 6 is the analysis of variance table for the content of volatile fatty acids. The quadratic polynomial regression equation of the relationship between the volatile fatty acid content and the ultrasonic power and ultrasonic time is shown in Eq. (5):

$$Y = 16.24 + 1.67A + 0.30B - 1.43AB - 2.77A^2 - 2.26B^2 \quad (5)$$

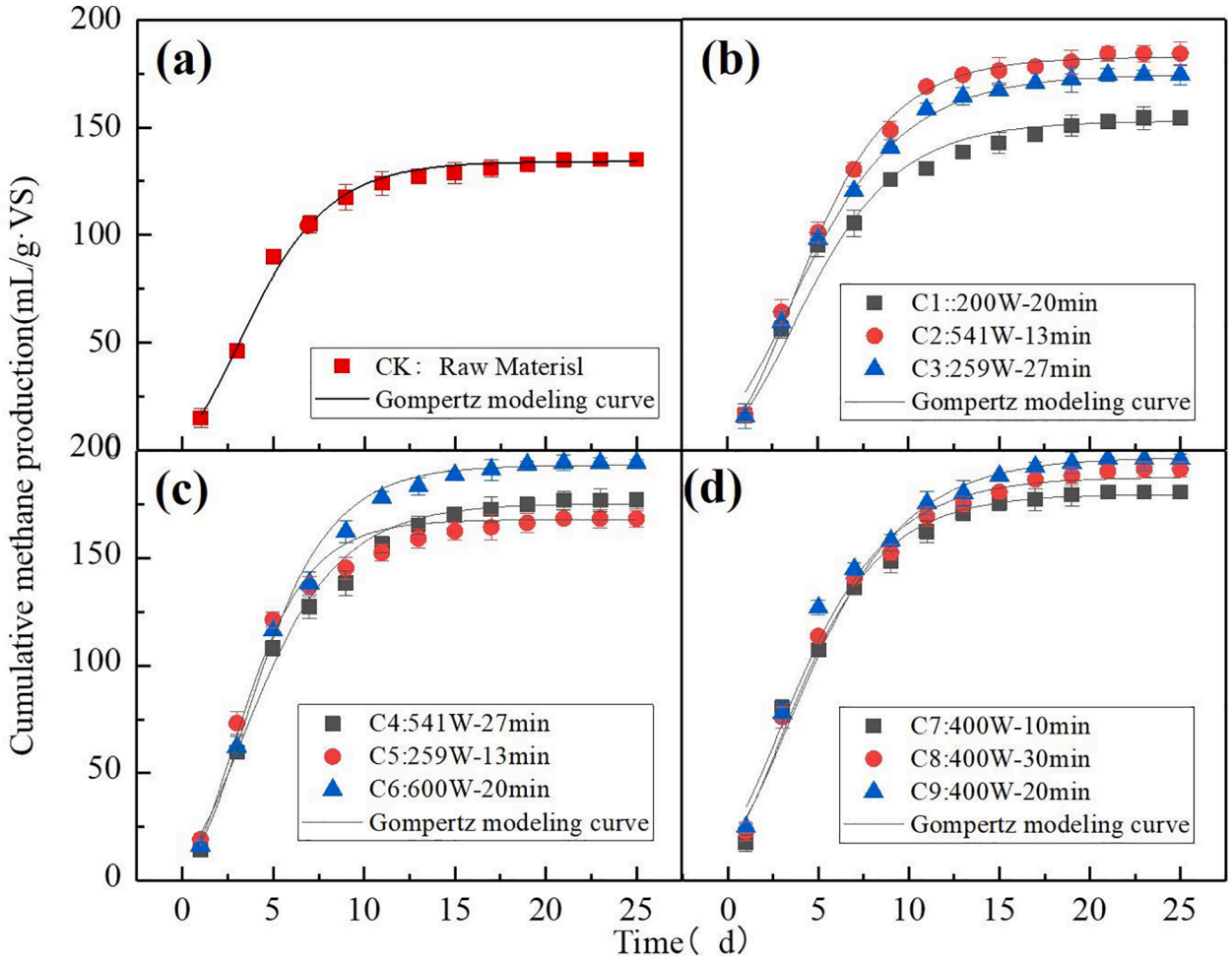


Fig. 3. Fitting results of cumulative methane production for different pretreatment conditions in anaerobic digestion。

As supplementary file shows that the F value of the quadratic model is 21.93 and the p value is 0.0040, indicating that the model is significant and practical analytical significance. *P*-value of AB is 0.0247, revealing that the interaction of two factors of ultrasonic power and time on VFAs is not significant. Due to the F-value of ultrasonic power is larger, it means that the effect of ultrasonic power on VFAs is greater than time, meanwhile, the effect of ultrasonic power on VFAs is significant ($p < 0.05$). For the effect of ultrasonic-aerobic hydrolysis pretreatment, aggravate the damage to CS, improve the degradation of lignocellulose and increase VFAs effectively. The time of ultrasonic-aerobic hydrolysis treatment is shorter than that time of aerobic hydrolysis. The results indicated that the introduction of ultrasonic waves can not only reduce the intensity of aerobic hydrolysis, short the time, but also increase the VFAs.

As shown in Fig. 2(a) and (b), within a certain range, VFAs will increase with the increase of the ultrasonic power and time. When the pretreatment conditions are 309 W and 26 min, VFAs reached the maximum value and no longer increases.Fig. 3

4. Conclusion

The study suggested that ultrasonic-aerobic hydrolysis pretreatment mainly changed the degradation lignocellulose rate and improved the methane production by increasing VFAs in slurry. Through RSM analysis, the effect of power on VFAs was obviously greater than that of time. The highest VFAs (16.24 g/L) was achieved by the pretreatment 309 W and 26 min, thereby methane production reached 198.56 mL/g VS, which was 46.73% higher than CK. Therefore, this study recommended ultrasonic-aerobic hydrolysis application for shortened the time and hydrolysis of CS effectively. Finally, using the potential to handle bulk amount of CS can be helpful in industrial applications.

CRediT authorship contribution statement

Xiaodong Chu: Conceptualization, Supervision, Writing – original draft, Funding acquisition, Project administration. **Qiushuang Cheng:** Supervision, Conceptualization, Data curation. **Yonghua Xu:** Supervision. **Lina Luo:** Conceptualization. **Ming Wang:** Supervision. **Guoxiang Zheng:** Supervision, Validation. **Hongqiong Zhang:** Formal analysis. **Weiming Yi:** Supervision, Validation, Visualization, Funding acquisition. **Xiaofeng Liu:** Conceptualization, Supervision, Funding acquisition. **Yongming Sun:** Conceptualization, Funding acquisition. **Yong Sun:** Conceptualization, Supervision, Writing – original draft, Funding acquisition, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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